

Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Distinguish confounding from mediation and from effect modification.
- Read a multiple regression coefficient as an *adjusted* effect.
- Centre predictors before fitting interactions and explain why.

Prerequisites

Sessions 1 and 2 of this week.

Background

Multiple regression is the standard way to estimate the effect of one predictor while holding others constant. The word *holding* is doing a lot of work: the adjusted coefficient is the effect conditional on the other variables in the model, not the effect one would observe in a new study that intervened on them. Confounding, mediation, and collider bias all pivot on which covariates are in the model. The statistics textbook cannot answer that question; the subject-matter science must.

Interactions complicate this picture. When two predictors interact, the effect of one depends on the level of the other, and the main-effect coefficient becomes the effect when the interacting variable is zero. Centring continuous predictors before fitting an interaction restores a readable interpretation: the main effect becomes the effect at the sample mean of the other variable, which is usually what the reader wants.

Collinearity and scale are often discussed together. Centring does not reduce *real* collinearity between X and Z, but it does reduce the algebraic collinearity between X and $X \cdot Z$ that otherwise makes the coefficients hard to interpret and inflates their standard errors.

Setup

```
library(tidyverse)
library(broom)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

Simulate a scenario in which a confounder masks the effect of interest, then add an interaction to show effect modification.

2. Visualise

```
age <- rnorm(n, 55, 10)
# smoker more common among older people in this simulation (confounding)
smoker <- rbinom(n, 1, plogis(-3 + 0.05 * age))
# outcome depends on both; smoker effect is stronger at older ages (interaction)
y <- 120 + 0.6 * age + 5 * smoker + 0.2 * smoker * (age - mean(age)) +
  rnorm(n, 0, 8)
dat <- tibble(age, smoker = factor(smoker, labels = c("no", "yes")), y)

ggplot(dat, aes(age, y, colour = smoker)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "Age (years)", y = "Outcome", colour = "Smoker")
```

Both lines climb with age; the smoker line sits higher and may slope more steeply.

3. Assumptions

The usual linear-model assumptions, plus an implicit assumption that all relevant confounders are in the model.

```
par(mfrow = c(2, 2))
plot(fit_full)
par(mfrow = c(1, 1))
```

4. Conduct

Crude smoker effect, then adjusted for age, then with interaction:

```
adjusted <- lm(y ~ smoker + age, data = dat)
inter <- lm(y ~ smoker * age, data = dat)

bind_rows(
  tidy(crude, conf.int = TRUE) |> mutate(model = "crude"),
  tidy(adjusted, conf.int = TRUE) |> mutate(model = "adjusted"),
  tidy(inter, conf.int = TRUE) |> mutate(model = "interaction")
) |>
  filter(term != "(Intercept)") |>
  select(model, term, estimate, conf.low, conf.high, p.value)
```

Now with centred age, which makes the `smokeryes` coefficient the smoker effect *at mean age*:

```
inter_c <- lm(y ~ smoker * age_c, data = dat_c)
tidy(inter_c, conf.int = TRUE)
```

5. Concluding statement

After adjusting for age, the estimated smoker-effect at the sample mean age was `round(coef(inter_c)["smokeryes"], 1)` units (95% CI: `round(confint(inter_c)["smokeryes", 1], 1)` to `round(confint(inter_c)["smokeryes", 2], 1)`), with evidence of effect modification by age (interaction coefficient `round(coef(inter_c)["smokeryes:age_c"], 2)`).

Common pitfalls

- Reading an adjusted coefficient as a population-level causal effect without thinking about what the adjustment set represents.
- Fitting an interaction without centring and then trying to interpret the main effects.
- Dropping covariates that look non-significant one at a time.

Further reading

- Greenland S (1998), *Confounding and collapsibility in causal inference*.
- VanderWeele TJ. *Explanation in Causal Inference*.
- Fox J. *Applied Regression Analysis and Generalized Linear Models*.

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)